

The general multiplicity theorem, all-equal case:

$$Z_d(n) \sim \frac{S_{1/2}(a)}{2^d(d-1)!} n^{-1/2}(\log n)^{d-1} \text{ for every } d$$

Claude, with Daniel Murfet

2026-09-06

Abstract

The $d = 2$ and $d = 3$ instances (units 35, 38) are promoted to a theorem for every dimension: for $k = (1, \dots, 1)$, $h = (0, \dots, 0)$ — all d candidate exponents equal to $\frac{1}{2}$, multiplicity d — the chart standard integral satisfies

$$Z_d(n) \sim \frac{S_{1/2}(a)}{2^d(d-1)!} n^{-1/2}(\log n)^{d-1} \quad (d \geq 2),$$

the $(\log n)^{m-1}$ of `thm:TaylorTree` with $m = d$. Three files: the iterated logarithmic primitives H_m , their inductive squeeze $|H_{k+1}(L) - A \log^{k+1} L / (k+1)!| \leq C_{k+1}(1 + \log L)^k$, and the d -fold chart recursion with its reduction $Z_d(n) = n^{-1/2} H_{d-1}(\sqrt{n} b^d)$. Zero `sorry/axiom`.

1 Setting

Each coinciding candidate exponent contributes one “divide by u and substitute” step, i.e. one more $\int dx/x$ applied to the previous primitive. Making this an induction requires (i) a family H_m closed under the operation $G \mapsto \int_0^x G(t)/t dt$ with uniform bounds, (ii) a squeeze whose error term is a polynomial in $\log L$ of degree one less than the main term, and (iii) a chart recursion $Z_{d+1}(c) = \int_0^b Z_d(cu) du$ whose reduction to H_d is the same identity at every level.

2 The things formalised

```

noncomputable def iterLogPrim ( a : ) : → →
  | 0 => quadPrimitive a
  | m + 1 => fun x => t in (0:)..x, iterLogPrim a m t / t
theorem integral_iterLogPrim_div_eq ... :
  ( u in Ioc 0 b, iterLogPrim a m (c * u) / u) = iterLogPrim a (m + 1) (c * b)
theorem iterLogPrim_sub_le ( a : ) (h : 0 < ) (k : ) (L : ) (hL : 1 ≤ L) :
  | iterLogPrim a (k+1) L - quadMass a * Real.log L ^ (k+1) / ((k+1).factorial : ) |
    iterConst a (k+1) * (1 + Real.log L) ^ k
noncomputable def iterChart ( a b : ) : → →
  | 0 => fun c => quadKernel a c
  | d + 1 => fun c => u in Ioc 0 b, iterChart a b d (c * u)
theorem iterChart_succ_eq ... :
  iterChart a b (d+1) c = c ^ 1 * iterLogPrim a d (c * b ^ (d+1))
theorem iterChart_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) (k : ) :

```

```

(fun n => iterChart a b (k+2) (Real.sqrt n)) ~[atTop]
  fun n => fluctuation (1/2) a / (2 ^ (k+2) * ((k+1).factorial : ))
    * (n ^ (-(1/2 : )) * Real.log n ^ (k+1))

```

3 Proof strategy

H_m is defined by interval integrals, so continuity is `continuous_primitive` once the integrand $H_{m-1}(t)/t$ is shown bounded; the bound $|H_m(t)| \leq E|t|$ on all of $\mathbb{R}$ is proved simultaneously with continuity by induction. The squeeze induction integrates the previous estimate against dx/x over $[1, L]$: the main term integrates exactly to $A \log^{k+2} L / (k+2)!$ ($\int_1^L \log^{k+1} x / x = \log^{k+2} L / (k+2)$ by FTC), the remainder is bounded by $C_{k+1} \int_1^L (1 + \log x)^k / x = C_{k+1} ((1 + \log L)^{k+1} - 1) / (k+1)$, and the piece on $[0, 1]$ by E ; with $C_{k+2} = E + C_{k+1} / (k+1)$ and $(1 + \log L)^{k+1} \geq 1$ the bound closes. The reduction $Z_{d+1}(c) = c^{-1} H_d(c b^{d+1})$ is an induction using the divide-and-substitute identity. For the asymptotic, rather than expanding $(\frac{1}{2} \log n + d \log b)^{k+1}$ with explicit polynomial bounds, the `IsEquivalent` calculus is used: $H_{k+1}(L) \sim A \log^{k+1} L / (k+1)!$ (from the squeeze, since $(1 + \log L)^k = o(\log^{k+1} L)$), composed with $L(n) = \sqrt{n} b^d \rightarrow \infty$; $\log L(n) \sim \frac{1}{2} \log n$ (a constant is $o(\log n)$); `.pow`, `.mul` with $n^{-1/2}$, and two `congrs`.

4 Roadblocks and resolutions

- Pi-typed operations: `IsEquivalent.mul/.pow` produce $f * g$ and $f \wedge n$ as functions; the final `congr_left/congr_right` steps convert them to λ -forms with `Pi.mul_apply` (`exact` also accepts the `defeq` directly).
- `HasDerivAt.pow` on `const + log` leaves a `Pi.add_apply` redex and a $0 * x + 1$; add both to the `simp` only before `field_simp`.
- `pow_nonneg` (by `linarith`) k inside a `rw` elaborates the tactic block before the base is known; ascribe the inequality's type.
- `field_simp` does not simplify $\log L \cdot (\log L)^{-1}$ inside a power nor $(1/2)^k \cdot 2^k$; rewrite $1 + \ell^{-1} = (1 + \ell)/\ell$ and expand $(\frac{1}{2} \log n)^{k+1}$, $2^{k+2} = 2^{k+1} \cdot 2$ first.
- The equation lemma for a recursive constant must be stated in the syntactic form the goal uses (`m + 1 + 1`, not `m + 2`) for `rw` to fire; `rfl` proves either.

5 What was Mathlib, what was new

Mathlib: `intervalIntegral.continuous_primitive`, `integral_add_adjacent_intervals`, `abs_integral_le_integral_abs`, `integral_mono_on`, `integral_eq_sub_of_hasDerivAt`, `IsEquivalent.comp_tendsto/.mul/.pow/.trans`, `isLittle0_const_left`, `tendsto_sqrt_atTop`, `Tendsto.atTop_mul_const`. New: the iterated logarithmic primitives with their uniform squeeze and the general- d multiplicity theorem in the all-equal case.

6 Lessons

When a family of asymptotic instances shares one inductive mechanism, formalise the mechanism (here: one uniform squeeze propagated through $\int dx/x$) and let the instances become corollaries;

and prefer the `IsEquivalent` calculus over explicit polynomial-in-log bounds for assembling final asymptotics.

7 Outcome

Across units 35–39 the multiplicity structure of `thm:TaylorTree` is now formalised in the simplest chart geometry: no logarithm when candidate exponents differ (unit 37), and $(\log n)^{d-1}$ with the exact constant $S_{1/2}(a)/(2^d(d-1)!)$ when all d coincide (this unit). What remains is the general (k, h) multivariate case with partially coinciding exponents, which needs the weighted primitives F_γ of the GPT-6 Astra plan.